ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** acidic	Acid turns blue litmus red and red remains red.
2	**(a)** 6.0 V	4 x 1.5.
3	**(c)** metal	Metal properties.
4	**(d)** burning of coal, ripening of fruit	Both produce new substances.
5	**(b)** normal	Girls often grow earlier.
6	**(a)** copper	Best conductor.
7	**(c)** 60 km/h	90/1.5.
8	**(b)** mouth, oesophagus, stomach, small intestine, large intestine	Standard order.
9	**(b)** the sunlight leaf	Starch made.
10	**(d)** 80°	i + r = 40 + 40.
11	**(b)** have continuous daylight	No change in facing.
12	**(d)** NaOH	NaOH.
13	**(d)** using damaged wires	Damaged wires are dangerous.
14	**(b)** red litmus blue	Basic.
15	**(b)** no moisture	Moisture is needed.
16	**(a)** growth spurt	Both grow taller.
17	**(d)** lower than over sea	Warm air rises, low pressure.
18	**(b)** 50 km/h	100/2.
19	**(d)** 960	16 x 60.
20	**(d)** transpiration	Water vapour condenses.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	A base (1); Soap solution (1).	1+1
22	Coin is a conductor, completing the circuit (1); Plastic is an insulator (1).	1+1
23	Silver reacts with sulphur compounds in air (1); Forms black silver sulphide (1).	1+1
24	Melting of wax is physical (1); Burning of wax vapour is chemical (1).	1+1
25	Larynx grows more in boys (1); Testosterone causes this (1).	1+1
26	Metals are good conductors of heat (1); Wood is an insulator (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	X: basic (1); Y: acidic (1); Reason from colours (1).	1+1+1
28	Total voltage 6 V (1); Matches bulb rating (1); Bulb glows normally (1).	1+1+1
29	No reaction (1); Copper is less reactive (1); It cannot displace iron (1).	1+1+1
30	Nail turns brown (1); Solution turns pale green (1); Chemical change (1).	1+1+1
31	Obesity risk (1); Low vitamins and minerals (1); Add fruits, vegetables, milk (1).	1+1+1
	OR: Radiation to nearby objects (1); Convection moves warm air (1); Conduction in the heating element (1).	1+1+1
32	40 min = 2/3 h (1); 12 / (2/3) = 18 km/h (1); 5 m/s (1).	1+1+1
	OR: Mouth: starch begins digestion (1); Stomach and small intestine: digestion completed (1); Large intestine: water removed, undigested egested (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Test with blue and red litmus (1); Two turn blue litmus red: vinegar, lemon (1); Two turn red litmus blue: lime water, soap (1); Sugar: no change (1); Do not taste the liquids (1).	1+1+1+1+1
34	Cell, bulb, wires, free ends (1); Place samples: iron nail, coin, eraser, plastic comb, graphite (1); Bulb glows for conductors (1); Table of results (1); Conclude metals and graphite conduct (1).	1+1+1+1+1
35	Collect samples (1); Observe lustre (1); Test conduction (1); Strike and listen (1); Table and conclusion (1).	1+1+1+1+1
36	Two beakers: tap water and salt water (1); Same nails and exposure (1); Observe for a week (1); Salt water nail rusts more (1); Salt speeds rusting (1).	1+1+1+1+1
	OR: Boys: testosterone (1); Voice, beard (1); Girls: oestrogen (1); Breast, periods (1); Both: growth spurt (1).	1+1+1+1+1
37	Conduction: wax pins on rod (1); Convection: permanganate crystals in water (1); Radiation: hand near a lamp (1); Observations noted (1); Conclusions drawn (1).	1+1+1+1+1
	OR: Total time 4 h (1); Average 240/4 = 60 km/h (1); Running time 3.5 h (1); 240/3.5 = 68.6 km/h (1); Running speed is higher (1).	1+1+1+1+1
38	Mouth: amylase on starch (1); Stomach: pepsin on proteins (1); Small intestine: bile for fat, pancreatic juice for all (1); End products: sugar, amino acids, fatty acids (1); Absorbed by villi (1).	1+1+1+1+1
	OR: Broad flat surface (1); Thin (1); Stomata for gases (1); Veins for transport (1); Chloroplasts with chlorophyll (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** 20 m	Ratio 2:1.
2	**(c)** evening sky, in the west	Soon after sunset.
3	**(c)** it contains acetic acid	Acid.
4	**(c)** becomes brighter	More voltage.
5	**(a)** a protective oxide layer forms	Al2O3 layer.
6	**(d)** crystals form when a hot saturated solution cools	Cooling.
7	**(c)** neck	Front of the neck.
8	**(c)** white loose cotton	Reflect and breathe.
9	**(a)** speed	Rise over run.
10	**(d)** four chambers	Mammal.
11	**(a)** taking food from its host	Parasite.
12	**(d)** half	Inversely.
13	**(c)** umbra	Total.
14	**(c)** basic	Typical of bases.
15	**(b)** 0 V	They cancel.
16	**(a)** non-metal	Non-metal.
17	**(c)** freezing of water, evaporation of water	States change only.
18	**(a)** larynx is growing	Voice box.
19	**(a)** radiation	Air is a poor conductor.
20	**(c)** 9 km/h	1.5 x 6.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	150/2.5 = 60 km/h (1); 60 x 5/18 = 16.7 m/s (approx.) (1).	1+1
22	Muscles need more energy and oxygen (1); More carbon dioxide has to be removed (1).	1+1
23	More light for photosynthesis (1); Greater food production (1).	1+1
24	Reflection 35° (1); Between rays 70° (1).	1+1
25	Axis is tilted (1); Sun above horizon for different durations (1).	1+1
26	$H_2SO_4 + 2\,NaOH \longrightarrow Na_2SO_4 + 2\,H_2O$ (1); Salt: sodium sulphate (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Plant in a closed jar with KOH (1); Control with water (1); Test leaves for starch: only control positive (1).	1+1+1
28	Incidence = 90 - 70 = 20° (1); Reflection 20° (1); Between rays 40° (1).	1+1+1
29	New moon aligned with Sun and Earth (1); Moon's umbra falls on Earth (1); Region in umbra sees total eclipse (1).	1+1+1
30	Both contain bases (1); Neutralise stomach acid (1); Rate of action differs (1).	1+1+1
31	Senses large current (1); Switch trips (1); Can be reset (1).	1+1+1
	OR: Coating is basic copper carbonate or oxide (1); Acid in lemon reacts (1); Surface becomes shiny (1).	1+1+1
32	Plants make glucose (1); Uses carbon dioxide and water (1); New substances formed (1).	1+1+1
	OR: Peer pressure (1); Curiosity (1); Awareness and refusal skills (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Normal drawn (1); Angle of incidence 30° (1); Angle of reflection 30° (1); Angle between incident and reflected 60° (1); Reflected ray on opposite side of the normal (1).	1+1+1+1+1
34	21 June: north summer (1); 22 Dec: north winter (1); Equinoxes: equal day and night (1); South opposite (1); Tilt same direction (1).	1+1+1+1+1
35	Fizzing (1); Gas released (1); Carbon dioxide (1); Acid + base reaction (1); Lime water turns milky with the gas (1).	1+1+1+1+1
36	Switch off main supply (1); Use insulated tools (1); Wear rubber gloves (1); Check with a tester (1); Stand on dry insulated mat (1).	1+1+1+1+1
	OR: Hold with tongs (1); Do not look directly (1); Collect the ash (1); Dissolve in water (1); Test with litmus (1).	1+1+1+1+1
37	Useful: cooking (1); Useful: making curd (1); Useful: photosynthesis (1); Harmful: rusting (1); Harmful: food spoiling (1).	1+1+1+1+1
	OR: Mood swings (1); Need for acceptance (1); Talk to family (1); Hobbies (1); Seek help if needed (1).	1+1+1+1+1
38	Black pipes absorb (1); Glass cover traps heat (1); Water circulates by convection (1); Hot water tank insulated (1); Saves fuel (1).	1+1+1+1+1
	OR: Friction and air resistance (1); Amplitude decreases (1); Spring or weights give energy (1); Regular winding (1); Clock continues (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** starch is delayed	Amylase missing.
2	**(b)** remain brown	No chlorophyll.
3	**(a)** 65°	90 - 25.
4	**(a)** no seasons	Same sunlight all year.
5	**(d)** sodium sulphate	Na_2SO_4.
6	**(a)** are insulators	Block current.
7	**(c)** blue litmus red	Acid.
8	**(c)** no air reaches it	Oil stops oxygen.
9	**(a)** fast growth needs more energy	Energy.
10	**(c)** sea	Sea is warmer at night.
11	**(a)** 2 h	60/30.
12	**(c)** increases	More oxygen needed.
13	**(b)** guard cells	Guard cells swell.
14	**(a)** noon	Sun overhead.
15	**(b)** sunset	Opposite the Sun.
16	**(b)** it is a base	Base.
17	**(b)** a cell, bulb and wires	Must have a source and indicator.
18	**(c)** it is lustrous	Lustre.
19	**(a)** to remove insoluble impurities	Clear solution.
20	**(a)** abdomen	Near the stomach.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Voltages cancel (1); Bulb does not glow (1).	1+1
22	Magnesium hydroxide as antacid (1); Calcium oxide for neutralising acid (1).	1+1
23	Solubility decreases with temperature fall (1); Extra solute separates (1).	1+1
24	Strong muscles and bones (1); Better mood (1).	1+1
25	Copper is a good conductor (1); Even and fast heating (1).	1+1
26	Effort and gravity act against uphill (1); Speed changes so motion is non-uniform (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Metal conducts (1); Plastic insulates (1); Wool reduces heat loss (1).	1+1+1
28	Going 3/5 = 0.6 h (1); Returning 3/3 = 1 h (1); Total 1.6 h (1).	1+1+1
29	Damages lungs (1); Causes cough and cancer (1); Reduces stamina (1).	1+1+1
30	Guard cells swell and the pore opens (1); They shrink and it closes (1); Light and water matter (1).	1+1+1
31	Shows the road ahead (1); Helps avoid accidents (1); Wide view (1).	1+1+1
	OR: Near the poles (1); Sun above horizon at midnight (1); During summer (1).	1+1+1
32	Avoid confusion (1); Safety (1); Correct use (1).	1+1+1
	OR: Switch off (1); Hold the plug, not the wire (1); Dry hands (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	Humans: lungs (1); Fish: gills (1); Insects: tracheae (1); Earthworm: moist skin (1); Amoeba: surface of the body (1).	1+1+1+1+1
34	Water the plant (1); Cover branch with a clear bag (1); Seal the stem (1); Place in sunlight (1); Droplets inside show transpiration (1).	1+1+1+1+1
35	Solar: Moon between Sun and Earth (1); Lunar: Earth between (1); Umbra and penumbra (1); Straight travel of light (1); Occurs rarely (1).	1+1+1+1+1
36	The Moon turns once on its axis in about 27 days (1); It also goes round the Earth in about 27 days (1); So the same face points towards the Earth (1); The far side is never seen from the Earth (1); The far side still gets sunlight (1).	1+1+1+1+1
	OR: Add litmus to acid: red (1); Add base drop by drop (1); Colour turns violet (1); $\mathrm{HCl + NaOH \longrightarrow NaCl + H_2O}$ (1); Evaporate to get salt (1).	1+1+1+1+1
37	Connect nichrome wire to cells (1); Pass current (1); Wire gets hot (1); Increase cells: hotter (1); Warn about burns (1).	1+1+1+1+1
	OR: $\mathrm{2\,Mg + O_2 \longrightarrow 2\,MgO}$ (1); $\mathrm{S + O_2 \longrightarrow SO_2}$ (1); Dissolve in water (1); MgO basic, SO2 acidic (1); Litmus test (1).	1+1+1+1+1
38	Fuel burns: chemical (1); Engine heats: physical (1); Paint protects (1); Rust forms: chemical (1); Exhaust gases (1).	1+1+1+1+1
	OR: Nutrition for growth (1); Sleep for rest (1); Hygiene for health (1); Exercise for fitness (1); All support mental health (1).	1+1+1+1+1

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